定理：设 $A \in C^{n\times n}$ 的**k**个相异的特征值为 $\lambda_1, \lambda_2 \cdots, \lambda_k$，则**A**为单纯矩阵的充要条件是存在唯一的一组幂等矩阵 $E_i \in C^{n\times n}, i = 1, 2, \cdots, \mathrm{k}$ 使

(1) $E_i E_j = 0 (i \neq j)$ (2) $\sum_{i=1}^{k} E_i = I$

(3) $A = \sum_{i=1}^{k} \lambda_i E_i$;

此处 $\lambda_i (1 \leq i \leq k)$ 称为**A**的谱值，$E_i \in C^{n\times n}, i = 1, 2, \cdots, \mathrm{k}$称为**A**谱阵，$A = \sum_{i=1}^{k} \lambda_i E_i$ 称为**A**的谱分解式。

证明必要性：

$$X^{-1}AX = diag\{\underbrace{\lambda_1,\cdots,\lambda_1}_{n_1},\cdots,\underbrace{\lambda_k,\cdots,\lambda_k}_{n_k}\} = \Lambda$$

$$X = (X_1,\cdots,X_k),\ \ X_i \in C_{n_i}^{n\times n_i}$$

$$X^{-1} = Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_k \end{pmatrix} \qquad Y_i \in C_{n_i}^{n_i\times n} \quad (i = 1,\cdots,k)$$

$$令 E_i = X_i Y_i \in C^{n\times n}$$

$$I = XX^{-1} = (X_1, \cdots, X_k)\begin{pmatrix} Y_1 \\ \vdots \\ Y_k \end{pmatrix} = \sum_{i=1}^{k} X_i Y_i = \sum_{i=1}^{k} E_i$$

$$I = X^{-1}X = \begin{pmatrix} Y_1 \\ \vdots \\ Y_k \end{pmatrix}\begin{pmatrix} X_1 & \cdots & X_k \end{pmatrix} = \begin{pmatrix} Y_1X_1 & \cdots & Y_1X_k \\ \cdots & \cdots & \cdots \\ Y_kX_1 & \cdots & Y_kX_k \end{pmatrix} = \mathrm{I}_n$$

所以 $Y_i X_i = I_{n_i}, \quad Y_i X_j = 0_{n_i \times n_j} \ (i \neq j)$

证明充分性：

取X_i为E_i的基构成的阵，则令

$$X=(X_1,\cdots,X_k)$$

1）证明X是方阵；2）证明X满秩

设$rank(E_i)=n_i\ \ (i=1,\cdots,k)$,则

$$X_i\in C_{n_i}^{n\times n_i}$$

且$n_i=rank(E_i)=tr(E_i)\ \ (i=1,\cdots k)$

$$\sum_{i=1}^{k}n_i=\sum_{i=1}^{k}trE_i=tr(\sum_{i=1}^{k}E_i)=trI_n=n$$

因此X是n阶方阵。

$$E_i = (\alpha_{i_1}, \cdots, \alpha_{i_n}) = (X_i\beta_1, \cdots, X_i\beta_n) = X_i(\beta_1, \cdots, \beta_n) = X_iY_i, \beta_n \in C^n$$

$$Y_i \in C^{n_i \times n} \qquad i = 1, \cdots, k$$

$$令 Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_k \end{pmatrix} \in \mathrm{C}^{n \times n}$$

$$所以 \qquad XY = (X_1 \cdots, X_k) \begin{pmatrix} Y_1 \\ \vdots \\ Y_k \end{pmatrix} = \sum_{i=1}^{k} X_iY_i = \sum_{i=1}^{k} E_i = I_n$$

因此X可逆。

$$AX = (\sum_{i=1}^{k} \lambda_i E_i)(X_1, \cdots, X_k) = ((\sum_{i=1}^{k} \lambda_i E_i) X_1, \cdots, (\sum_{i=1}^{k} \lambda_i E_i) X_k)$$

$$= (\lambda_1 E_1 X_1, \cdots, \lambda_k E_k X_k) = (\lambda_1 X_1, \cdots, \lambda_k X_k)$$

$$= (X_1, \cdots, X_k) \begin{pmatrix} \lambda_1 I_{n_1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n I_{n_k} \end{pmatrix} = X\Lambda$$

$$X^{-1}AX = diag\{\underbrace{\lambda_1, \cdots, \lambda_1}_{n_1}, \cdots, \underbrace{\lambda_k, \cdots, \lambda_k}_{n_k}\}$$

证明唯一性：

设$F_1,\cdots,F_k$满足 $F_i^2=F_i \quad (i=1,\cdots,k) \quad F_iF_j=0 \ (i\neq j)$

$$\sum_{i=1}^{k}F_i=I_n \qquad A=\sum_{i=1}^{k}\lambda_iF_i$$

则$AE_i=E_iA=\lambda_iE_i,\ AF_i=F_iA=\lambda_iF_i \qquad (1\le i\le k)$

$$AE_iF_j=\lambda_iE_iF_j=E_iAF_j=\lambda_jE_iF_j$$

因此，当$i\neq j$时 $E_iF_j=0$

$$E_i=E_iI_n=E_i(\sum_{j=1}^{k}F_j)=E_iF_i=(\sum_{j=1}^{k}E_j)F_i=I_nF_i=F_i \quad i=1,\cdots,k$$

例题

（1）设 $A=\begin{pmatrix}1 & 1\\ 4 & 1\end{pmatrix}$ 求矩阵A的谱分解

解

由 $f_A(\lambda)=\begin{vmatrix}\lambda-1 & -1\\ -4 & \lambda-1\end{vmatrix}=(\lambda+1)(\lambda-3)$

得 $\lambda_1=3,\lambda_2=-1,$

$$x_1=\begin{pmatrix}1\\ 2\end{pmatrix},x_2=\begin{pmatrix}1\\ -2\end{pmatrix}$$

$$X = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \qquad X^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & -\frac{1}{4} \end{pmatrix}$$

$$E_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ 1 & \frac{1}{2} \end{pmatrix} \qquad E_2 = ?$$

（2）求单纯矩阵 $A=\begin{pmatrix}1&2&2\\2&1&2\\2&2&1\end{pmatrix}$ 的谱分解

由矩阵A的特征多项式 $|\lambda E-A|=(\lambda+1)^2(\lambda-5)$

得A的特征值 $\lambda_1=-1,\lambda_2=5$ 及相应的线性无关的特征向量为 $x_1=(1,-1,0)^T, x_2=(1,0,-1)^T, x_3=(1,1,1)^T$

$$X^{-1}=\begin{pmatrix}\frac{1}{3}&-\frac{2}{3}&\frac{1}{3}\\\frac{1}{3}&\frac{1}{3}&-\frac{2}{3}\\\frac{1}{3}&\frac{1}{3}&\frac{1}{3}\end{pmatrix}$$

则

$$E_1 = (X_1, X_2)\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$E_2 = x_3 y_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

从而 $A = -E_1 + 5E_2$

（3）求正规矩阵 $A=\begin{pmatrix}0&1&1&-1\\1&0&-1&1\\1&-1&0&1\\-1&1&1&0\end{pmatrix}$ 的谱分解

由矩阵A的特征多项式 $|\lambda E-A|=(\lambda-1)^3(\lambda+3)$

得A的特征值 $\lambda_1=1$(三重), $\lambda_2=-3$

对于 $\lambda_1=1$, 相应的线性无关的特征向量为

$$x_1=(1,1,0,0)^T, x_2=(1,0,1,0)^T, x_3=(-1,0,0,1)^T$$

对于 $\lambda_2=-3$, 相应的特征向量为 $x_4=(-1,1,1,-1)^T$

将x_1,x_2,x_3标准正交化得：

$$\alpha_1=\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},0,0\right)^T,\alpha_2=\left(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}},0\right)^T,\alpha_3=-\left(\frac{1}{\sqrt{12}},\frac{1}{\sqrt{12}},\frac{1}{\sqrt{12}},\frac{3}{\sqrt{12}}\right)^T$$

将x_4标准化 $\alpha_4=\left(-\frac{1}{2},\frac{1}{2},\frac{1}{2},-\frac{1}{2}\right)^T$

记 $U_1=(\alpha_1,\alpha_2,\alpha_3),U_2=(\alpha_4)$

$$P_1=U_1U_1^H,P_2=U_2U_2^H$$

则 $A=\lambda_1P_1+\lambda_2P_2=P_1-3P_2$